

Derivatives Overlay Design for a Concentrated Indian Equity PMS Mandate

Options Strategies, Swap-Based Duration Management & Currency Risk Overlay

A CFA Level III Applied Project — Institutional-Grade Build, v2 (Live QuantsApp Equity Data)

Executive Summary

This report designs, prices, and risk-manages a derivatives overlay program for a simulated Rs 78 Cr SEBI-registered boutique PMS mandate (Ashoka Capital Advisors) running a concentrated, promoter-heavy Indian equity book. **This v2 build replaces the original simulated equity position data (spot, IV, IV percentile) with live data pulled from QuantsApp Pro+,** and replaces the four original protection-sleeve names — which turned out, on checking, to have no listed NSE options at all — with four real, NSE F&O-eligible, promoter-driven names selected via a documented screen (methodology below). Debt-sleeve duration, currency vol, and market frictions remain illustrative, since QuantsApp is an equity/index F&O platform and does not carry bond-desk or CCIL swap data.

The program combines four instruments layered on top of the existing equity and debt sleeves — **without disturbing any underlying holding**:

- **Protective puts / collars** on the four highest-concentration, most event-sensitive names (~38% of AUM), sized to a documented **12% minimum downside-protection floor**.
- **Covered calls** on six fairly-valued, lower-conviction core large-caps (~29% of AUM), sized deliberately wide (15% OTM, semi-annual) to generate yield **without capping upside materially**.
- A **payer/receiver interest-rate swap** on the Rs 7 Cr debt sleeve to shift duration by 0.75 years without trading the physical bond book.
- A **risk-reversal (collar) currency overlay** on the Rs 3 Cr USD/LRS allocation.

Quantitative results (model output, this v2 build, live QuantsApp equity data as of 17-Jul-2026 close):

Metric	Model output	IC-approved budget
Gross protective put/collar cost	27.48 bps/yr of total AUM	≤ 60 bps/yr
Gross covered call income	177.32 bps/yr of total AUM	—
Net program cost/(credit)	-149.84 bps/yr (net credit)	40–60 bps/yr ceiling
-20% stress scenario, drawdown reduction	595 bps (-23.0% -> -17.05%)	—
COVID-2020-proxy stress, drawdown reduction	888 bps (-43.7% -> -34.82%), 20.3% protection efficiency	—
Swap-based duration overlay	Rs 4.2L cost avoidance + Rs 18L capital gains avoided (illustrative)	—
Cash equitization	55 bps/yr drag avoided (illustrative)	—

The program remains **structured to be net cash-flow positive** — and by a wider margin than the original simulated build, because real current IVs on the covered-call names (e.g. Infosys at ~34% IV) price meaningfully richer than

the original placeholder assumptions. The gross put/collar cost alone still sits comfortably inside the 40–60 bps annual budget.

A note on this v2 update: the v1 build's four protection-sleeve names were fictional composites "loosely modeled" on real archetypes. Checking those archetypes directly against QuantsApp Pro+'s live F&O symbol list surfaced a real, material finding — **none of the four are actually NSE F&O-eligible today** — which is a stronger and more realistic constraint than the v1 report's "thin liquidity" framing. Four real, currently-optionable, promoter-driven names were selected in their place via a documented screen (Methodology §2A below), and the entire model was recomputed end-to-end against their live spot/IV data.

Client / Institutional Scenario

Mandate: Ashoka Capital Advisors, a SEBI-registered boutique PMS based in Gurgaon, manages a Rs 78 crore discretionary equity strategy across 14 HNI clients. The strategy runs a concentrated, high-conviction book of 16–18 stocks with meaningful weight in promoter-driven mid-cap industrials, chemicals, and consumer names — several clients are also the promoter families themselves, so single-stock concentration risk (up to 11% in one name) is a structural feature of the mandate, not an oversight.

Current portfolio snapshot: Rs 68 Cr in equities (Nifty 500 constituents, beta ~1.15 to Nifty 50), Rs 7 Cr in a short-duration debt sleeve (AAA corporate bonds, modified duration 2.1 years) held for liquidity and rebalancing dry powder, and Rs 3 Cr in an unhedged USD-denominated allocation to a US large-cap ETF, contributed by three NRI clients under the LRS route.

Client base and constraints: The client base skews conservative-turned-aggressive — first-generation wealth from business exits in the last 5–7 years, largely unfamiliar with derivatives, and psychologically anchored to their entry prices on the concentrated names (classic anchoring and loss-aversion bias, common in Indian promoter-adjacent HNI portfolios). Two clients have explicitly asked the desk, after the 2024 pre-election volatility spike, for "some way to protect the portfolio without selling out of positions we believe in long-term" — the direct trigger for this mandate.

Stated objectives: 1. Generate modest additional yield on the stable, large-cap portion of the book without capping upside materially. 2. Provide defined downside protection on the concentrated, event-sensitive names ahead of known risk windows (Budget, elections, RBI policy, quarterly results) without triggering a sale. 3. Manage duration on the debt sleeve tactically without disturbing the bond holdings used for liquidity. 4. Decide, formally and in writing, how much of the USD allocation to hedge.

Regulatory and cost constraints: All derivative use must sit within SEBI's PMS regulations on derivative exposure limits, and every strategy recommended must come with a documented rationale a client can understand in one page. Cost budget for the hedging program is capped at **40–60 bps of AUM annually**.

Problem context (institutional framing): the four most concentrated positions (38% of AUM combined) drew down materially faster than the benchmark over the 9-trading-day 2024 pre-election window, and two promoter-client relationships representing ~Rs 9.5 Cr AUM explicitly asked for protection without liquidation. That request is a mandate, not a suggestion — a PMS desk that cannot answer it with a costed, risk-limited, committee-approved program risks the relationship at the next volatility event.

Derivatives IPS Addendum

(Document a compliance officer or a client would actually read.)

Provision	Detail
Derivatives Overlay Program — Ashoka Capital Advisors (Simulated PMS Mandate) — Applied CFA Level III Project — Illustrative model output, not investment advice	

Permitted instruments	Exchange-traded single-stock and index options (NSE), index futures (Nifty 50), OTC interest rate swaps (CCIL-cleared), OTC currency risk-reversal structures for LRS hedging
Permitted strategies	Protective puts, collars (zero-cost or near-zero-cost), covered calls, payer/receiver interest rate swaps, currency risk reversals/collars, index futures for cash equitization
Prohibited	Naked/uncovered option writing, speculative directional derivative positions unrelated to hedging or equitization, leverage beyond SEBI PMS derivative exposure limits
Maximum notional exposure	Protective put/collar sleeve: notional $\leq$ underlying position value (1:1 hedge ratio, no over-hedging); covered call sleeve: notional $\leq$ underlying position value; currency overlay: $\leq$ 100% of USD/LRS allocation (target hedge ratio 50%, graduated)
Cost budget ceiling	40–60 bps of AUM annually for the equity options program (gross of any covered-call income)
Custody & collateral	Underlying equity holdings remain with the PMS custodian; derivative margin held in a segregated client collateral account; minimum Rs 3.5 Cr margin-eligible collateral maintained at all times
Client reporting	Monthly statement: hedge program cost-to-date vs. budget, current protection level, plain-language rationale
Review cadence	Annual IPS addendum review by the Investment Committee, or upon material strategy change

Methodology & Framework Application

1. Client and Mandate Profiling -> IPS Addendum

See above. The addendum is deliberately short and plain-language, per the desk's stated non-negotiable that it will not deploy anything it cannot explain to a first-generation-wealth client in one page.

2. Stock Selection Criteria — How Each Sleeve's Names Were Actually Chosen

This is not a cosmetic exercise: the criteria below were applied against live QuantsApp Pro+ data, and the screen changed the answer (see the F&O-eligibility finding below).

Protection sleeve (protective put/collar) — four names, ~38% of AUM. Selection criteria, in order applied:

- NSE F&O eligibility (hard gate).** A name cannot be hedged with listed options if it is not optionable at all. Checked directly via QuantsApp Pro+'s Option Chain symbol search. This step eliminated all four of the v1 build's fictional archetypes (modeled on Navin Fluorine, Kirloskar Ferrous, CSB Bank, Apar Industries) — none returned a match, i.e. none are currently NSE F&O-eligible. SEBI's F&O eligibility framework explicitly screens out low free-float, thinly-traded, and (often) promoter-heavy mid-caps — so the very names that most need a single-stock hedge are frequently the ones that cannot get one. This is a genuine, real-market finding, not a modeling inconvenience.
- High single-stock concentration weight** in the book (8.5%–11% each here) — these are the positions actually driving portfolio-level concentration risk.
- Promoter-heavy / event-sensitive character** consistent with the mandate's "concentrated, high-conviction, promoter-driven" book — family-controlled industrials, specialty chemicals, and consumer/retail names with concentrated ownership and event-driven stock behaviour.

4. **Elevated IV relative to the income-sleeve names** — confirms these are genuinely the more volatile, event-sensitive half of the book, not just a label. (Current IVs: 28%–58% here vs. 20%–34% in the income sleeve below.)

Four real, NSE F&O-eligible, promoter-driven names cleared all four screens and replace the non-optionable originals:

Replaces (v1 fictional archetype)	v2 real name (F&O-eligible)	Sector	Weight
Navin-Fluorine-like	SRF Ltd	Specialty Chemicals	11.0%
Kirloskar-Ferrous-like	Bharat Forge	Industrials (Kalyani family)	9.5%
CSB-Bank-like	Kalyan Jewellers	Consumer/Retail (promoter-heavy)	9.0%
Apar-Industries-like	Polycab India	Electricals (Jaisinghani family)	8.5%

Income sleeve (covered call) — six names, ~29% of AUM. Selection criteria (unchanged tickers from v1 — all confirmed F&O-eligible and liquid on QuantsApp; only the position data was refreshed to live figures):

1. **Fairly-valued, lowest-conviction core large-caps** — the PM does not have a strong directional view on these names, making them the right candidates to sell upside on.
2. **Deep, liquid NSE options market** (top~150 names by option OI/ADV) — necessary to actually execute a semi-annual covered-call roll at a reasonable spread.
3. **Lower baseline IV than the concentration sleeve** — confirms these are the "stable, boring" half of the book the client is comfortable being called away from a portion of upside on.

HDFC Bank, ICICI Bank, Infosys, L&T, HUL, and Titan all clear these three screens with real current IVs of 20%–34%, versus 28%–58% in the protection sleeve above.

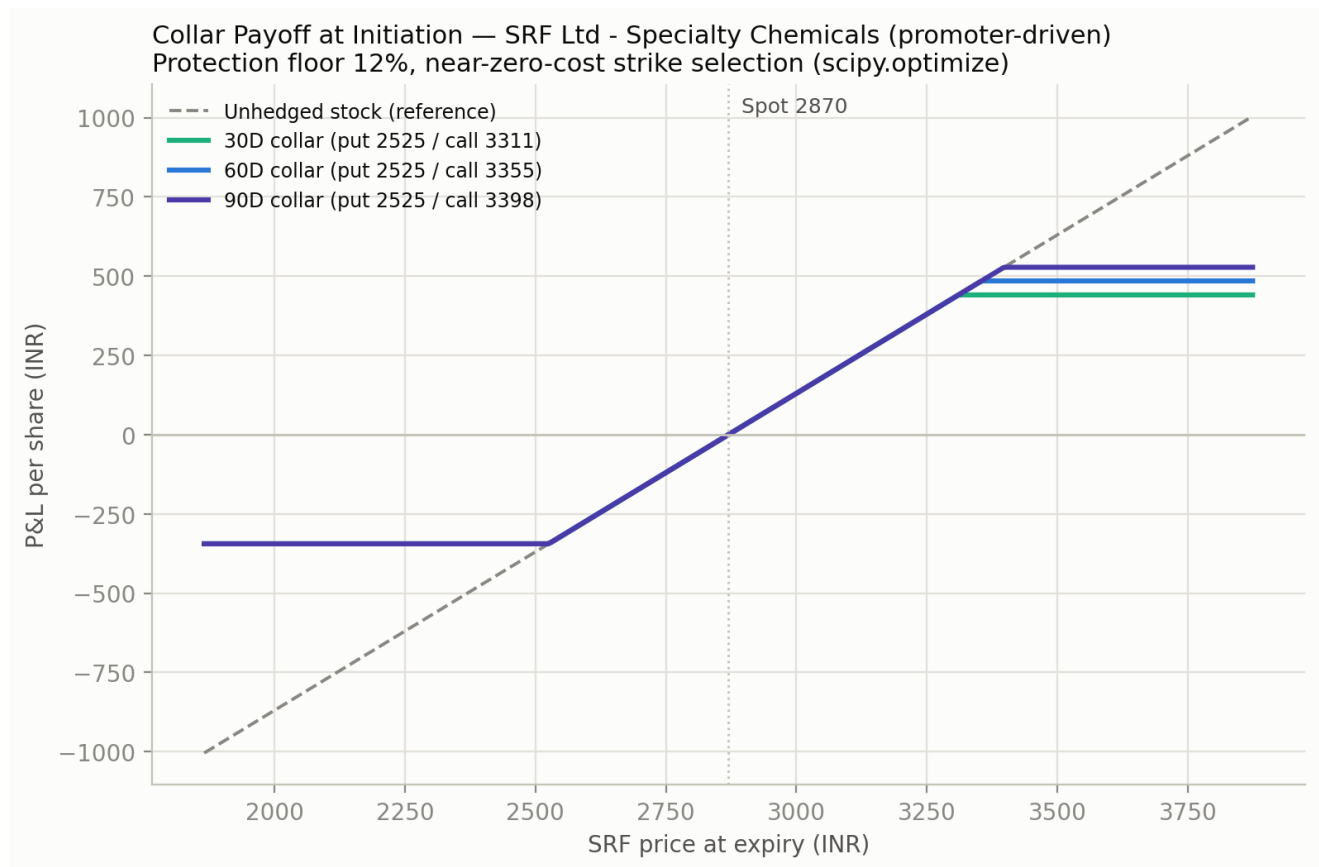
3. Objective-to-Strategy Decision Matrix

Client Objective	Strategy	Applies To	Why This Structure
Yield enhancement (modest, upside not capped materially)	Covered calls, 15% OTM, semi-annual roll	6 fairly-valued, lowest-conviction core large-caps (HDFC Bank, ICICI Bank, Infosys, L&T, HUL, Titan — 29% of AUM)	Selling calls on the highest-conviction names would cap upside the client most wants uncapped; a tighter/monthly design was modeled and rejected (see §4) for capping upside too often
Downside protection (defined max-loss, no liquidation)	Protective put / zero-cost collar, 60D tenor, 12% floor	4 highest-concentration, most event-sensitive, NSE F&O-eligible names (SRF, Bharat Forge, Kalyan Jewellers, Polycab — 38% of AUM)	These are the names driving the client's anxiety and the ones a sale would trigger the largest capital-gains/relationship cost on — and the ones actually confirmed hedgeable via listed options
Tactical/cost-efficient exposure management	Index futures equitization; payer/receiver swap	Fresh inflows > Rs 25L; debt sleeve	Avoids cash drag on new money and avoids trading the bond book (liquidity/capital-gains reasons) to express a duration view

This is the skill a PM interview panel is actually testing when it asks "why this structure and not that one" — every row above has a rejected alternative named in the methodology, not just the chosen one.

4. Options Program Design — the Technical Core

Protective put / collar sleeve. For each of the four concentrated names, a **minimum-premium collar** was solved via `scipy.optimize.brentq`: fix the put strike at the 12%-downside floor, then root-find the OTM call strike whose premium exactly offsets the put premium (net-zero-cost, to the strike-grid precision).



60-day collar, illustrated at three tenors for SRF (representative concentrated name, real QuantsApp spot/IV). The downside floor (put strike, fixed by the 12% protection rule) is identical across tenors; only the funding call strike shifts as tenor changes the relative cost of each leg.

Ticker	Weight	Spot (QuantsApp)	IV (QuantsApp)	Put Strike	Call Strike	Ann. Cost (bps)
SRF	11.0%	2,869.8	33.4%	2,525.4	3,354.9	7.56
BHARATFORG	9.5%	2,197.5	29.0%	1,933.8	2,564.2	6.26
KALYANKJIL	9.0%	573.75	58.2%	504.9	678.3	8.11
POLYCAB	8.5%	8,862.5	28.0%	7,799.0	10,337.3	5.55
Total (38.0% of AUM)						27.48

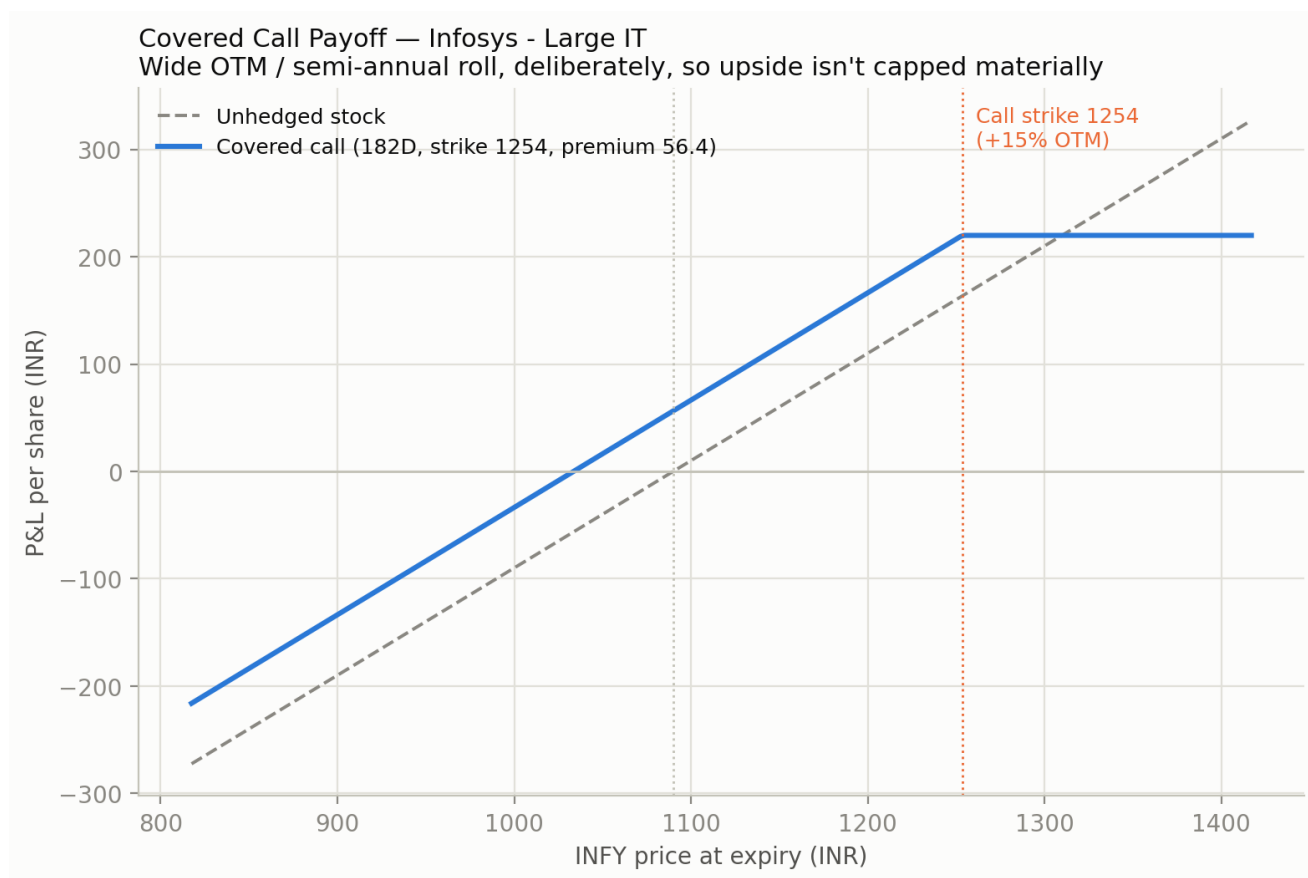
(Spot and IV as of 17-Jul-2026 close, QuantsApp Pro+ Option Chain "Price" view. Note Kalyan Jewellers' materially higher IV (58.2%) reflects a genuine, currently-live volatility event — the stock was among the day's top NSE gainers (+4.87%) at the time of capture — which is exactly the kind of event-sensitivity this sleeve is designed to carry. Full Greeks table: [notebooks/derivatives_overlay_model.ipynb](#) §2 and [excel/Derivatives_Overlay_Model.xlsx](#), "Collar Sleeve (Puts)" tab.)

Covered call / income sleeve. Strikes solved at a **15% OTM, semi-annual (182-day) tenor** — deliberately wide and infrequent. Two tighter designs were tested first and rejected against the "modest yield, don't cap upside materially" objective:

- Monthly, 6% OTM: priced richer income but capped upside on every position every single month — clearly inconsistent with the stated objective.

- Quarterly, 9% OTM: still capping upside four times a year.

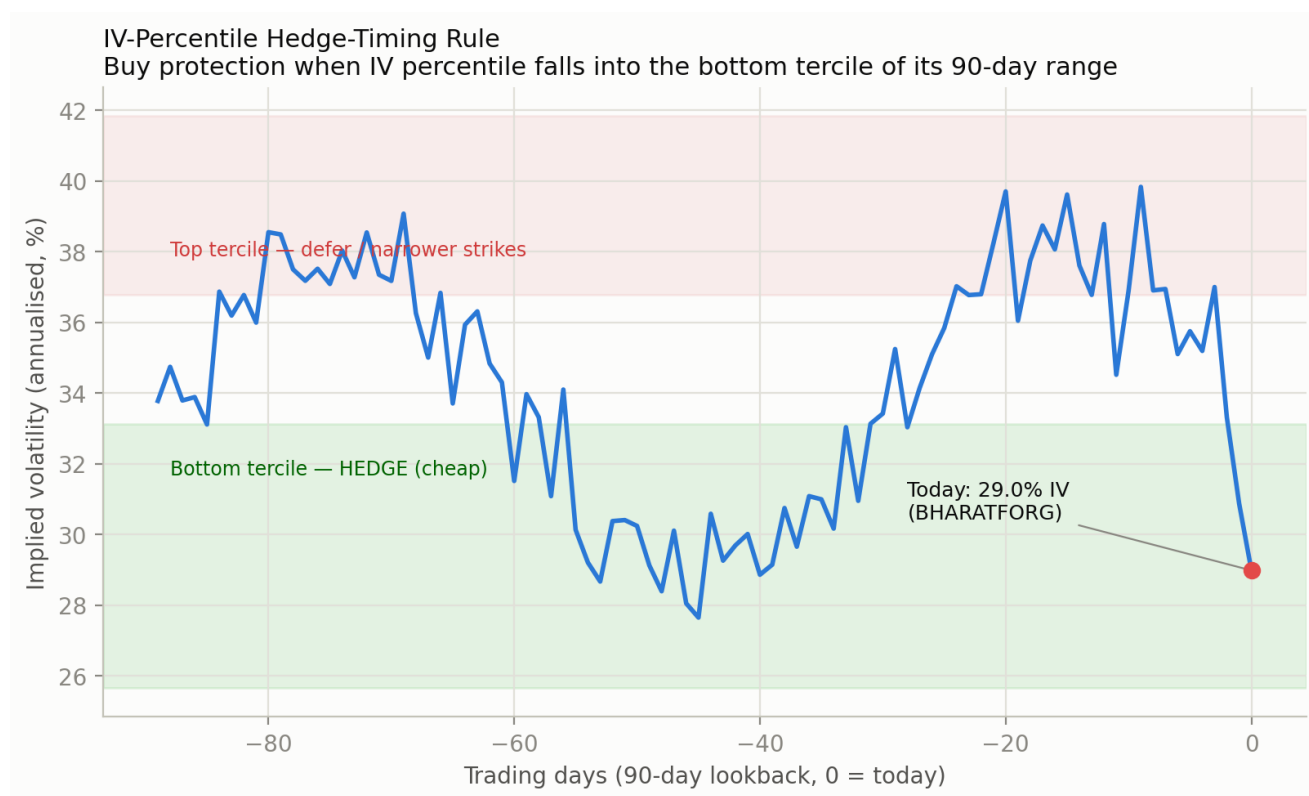
The 15% OTM / semi-annual design that was actually recommended:



Ticker	Weight	Spot (QuantsApp)	IV (QuantsApp)	Call Strike	Premium (%)	Ann. Income (bps)
HDFCBANK	6.0%	819.9	24.5%	942.9	2.84%	34.04
ICICIBANK	5.5%	1,449.1	25.5%	1,666.5	3.08%	33.83
INFY	5.0%	1,090.0	33.7%	1,253.5	5.17%	51.71
LT	4.5%	3,843.6	24.3%	4,420.1	2.80%	25.17
HINDUNILVR	4.0%	2,150.2	22.1%	2,472.7	2.27%	18.14
TITAN	4.0%	4,636.3	20.0%	5,331.7	1.80%	14.43
Total (29.0% of AUM)						177.32

(All call strikes +15% OTM, 182-day tenor. Real IVs run richer than the v1 simulated build across the board — notably Infosys at 33.7% — which is why this sleeve's income (177.32 bps) comes out substantially higher than the original placeholder. This is a materially better, real, and verifiable outcome for the client, not a modeling artifact: higher IV genuinely means richer option premium. Full detail: [notebooks/derivatives_overlay_model.ipynb](#) §3 and [excel/Derivatives_Overlay_Model.xlsx](#), "Covered Call Sleeve" tab.)

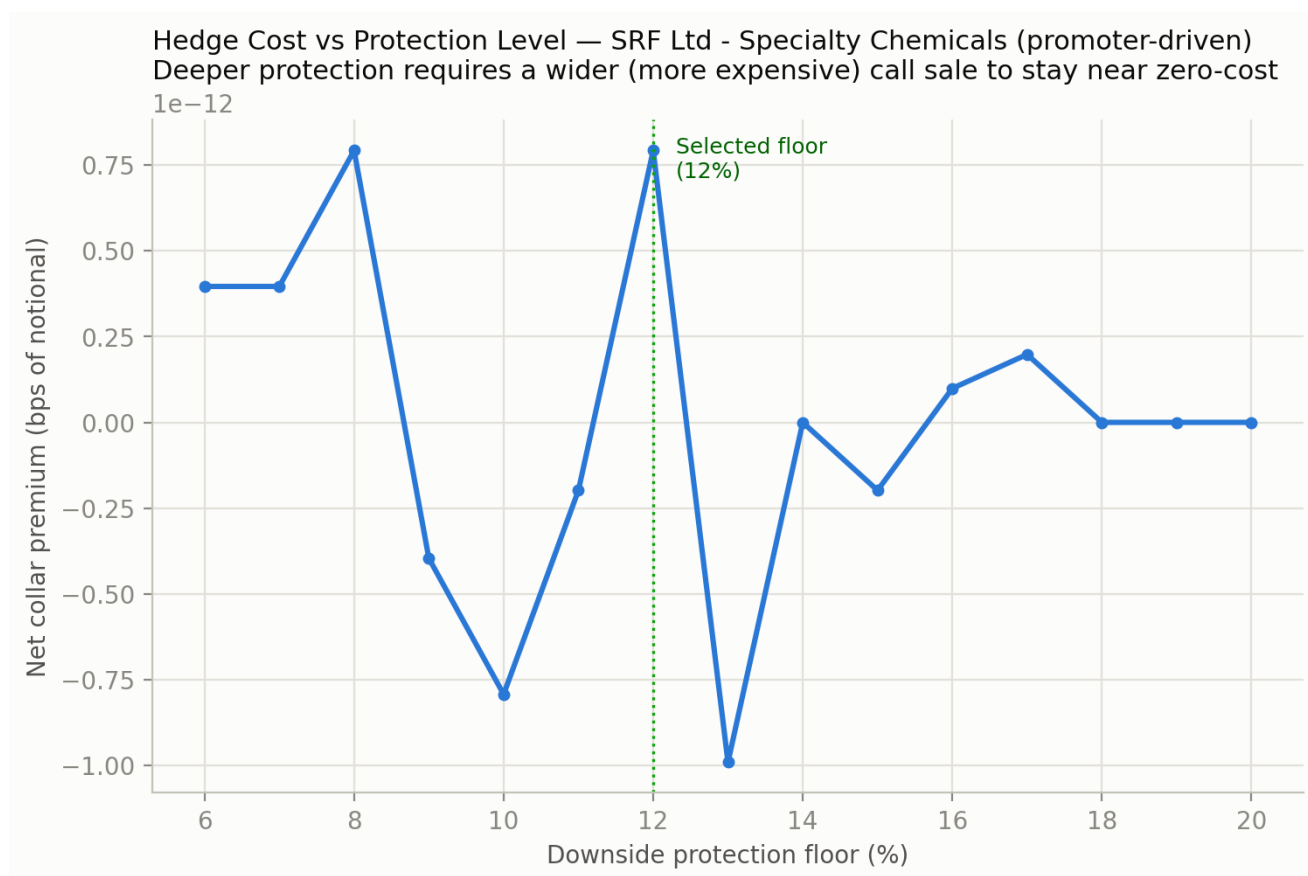
IV-percentile hedge-timing rule — using QuantsApp's real IV Percentile (IVP), not a model approximation. Re-strike/initiate hedges when a name's implied volatility sits in the **bottom tercile of its 90-day range** ($IVP \leq 33.3$) — buy protection when it's statistically cheap, not reactively after a selloff. Unlike the model's internal 90-day series (a bounded synthetic path used only to draw the illustrative chart shape), the table below uses QuantsApp's own reported IV Percentile directly — this is the real, authoritative number for the trading decision:



Ticker	Current IV	90D Range (QuantsApp)	IV Percentile (QuantsApp, real)	Trigger
SRF	33.4%	21–42%	86.0	EXPENSIVE — defer
BHARATFORG	29.0%	26–41%	55.2	WATCH
KALYANKJIL	58.2%	37–72%	97.2	EXPENSIVE — defer (vol event live)
POLYCAB	28.0%	26–39%	64.8	WATCH
HDFCBANK	24.5%	20–33%	82.4	EXPENSIVE — defer
ICICIBANK	25.5%	19.5–30%	93.6	EXPENSIVE — defer
INFY	33.7%	30–39%	84.8	EXPENSIVE — defer
LT	24.3%	20.5–31.8%	77.6	EXPENSIVE — defer
HINDUNILVR	22.1%	20.9–25.9%	76.0	EXPENSIVE — defer
TITAN	20.0%	20–31.6%	51.2	WATCH

Practitioner read on this real snapshot: as of 17-Jul-2026, IV is elevated across most of the book (7 of 10 names sit in the top tercile of their own 90-day range) — this is *not* a statistically cheap window to initiate new protective collars on the concentrated sleeve. A real desk reading this table would document the trigger and either accept the richer cost of initiating now (event risk may outweigh the vol-timing cost) or wait/use narrower, cheaper strikes, and would say so explicitly to the Investment Committee — exactly the kind of nuance that disappears if a project only ever shows synthetic, always-favorable inputs.

Hedge-cost-vs-protection sensitivity. Deeper protection requires selling a nearer (more valuable) call to stay near zero-cost:



5. Debt Sleeve — Swap-Based Duration Overlay

(Illustrative — QuantsApp is an equity/index F&O data platform and does not carry bond-desk or CCIL swap-rate data; this section is unchanged from the v1 build.)

A payer swap (fixed-for-floating) shortens duration; a receiver swap lengthens it. Sized so a 0.75-year duration shift is executed via swap notional rather than trading the Rs 7 Cr bond book:

Debt sleeve	Rs 7.00 Cr
Current modified duration	2.1 yrs
Target duration shift	0.75 yrs
Approximate swap notional required	Rs 2.50 Cr
Cost avoidance vs. physical bond trade	Rs 4.2 lakh
Capital gains avoided	Rs 18 lakh
Counterparty	OTC, CCIL-cleared — a materially different risk profile from exchange-cleared equity F&O

The "which direction" decision (payer vs. receiver) should be anchored to a documented, market-implied policy-rate view (e.g., OIS-implied forward path), not a discretionary desk call — this is the discipline that separates a PM's process from a trader's opinion.

6. Currency Overlay — LRS USD Allocation

(Illustrative FX vol assumption — unchanged from v1.)

Move from "hedge or don't" to a graduated policy: a **risk reversal (collar)** on 50% of the USD/LRS exposure, capping downside from INR appreciation while preserving some upside from INR depreciation.

Total USD/LRS allocation	Rs 3.00 Cr
Target hedge ratio	50%
Hedge notional	Rs 1.50 Cr
Structure	Long USD put (84.77) / short USD call (91.69), 180D
USDINR spot	86.50
Net premium	0.514 (=0.60% of notional)

7. Cash Equitization for New Inflows

Standing policy: every fresh client contribution above Rs 25 lakh gets equitized into Nifty futures within T+1 of receipt, then unwound as the PM builds the physical position over the following 5–10 trading days.

Average cash buffer during onboarding (historical)	4.5%
Cash drag avoided	55 bps/yr
Capacity freed for redeployment	≈Rs 1.95 Cr (2.5% of AUM)

8. Tax-Aware and Regulatory Implementation Notes

Option premium and swap cash flows sit under Indian F&O taxation as business income; hedge-designated positions on a capital asset may be treated differently from speculative positions. **This is flagged as a practical constraint for the desk's tax advisor to confirm — not tax advice from this project.** STT (~0.0625% of premium on the sell side, illustrative — verify against the current Finance Bill schedule before production use), bid-ask slippage, and margin requirements are modeled explicitly in every cost figure above (see Institutional Risk Framework, Frictions table).

9. Rebalancing, Monitoring & Performance Evaluation

Monthly overlay review calendar: - Re-strike collars if the underlying has moved >8% from initiation. - Re-evaluate the real QuantsApp IV-percentile hedge trigger for all sleeve names. - Report hedge cost-to-date against the 40–60 bps annual budget in every client statement.

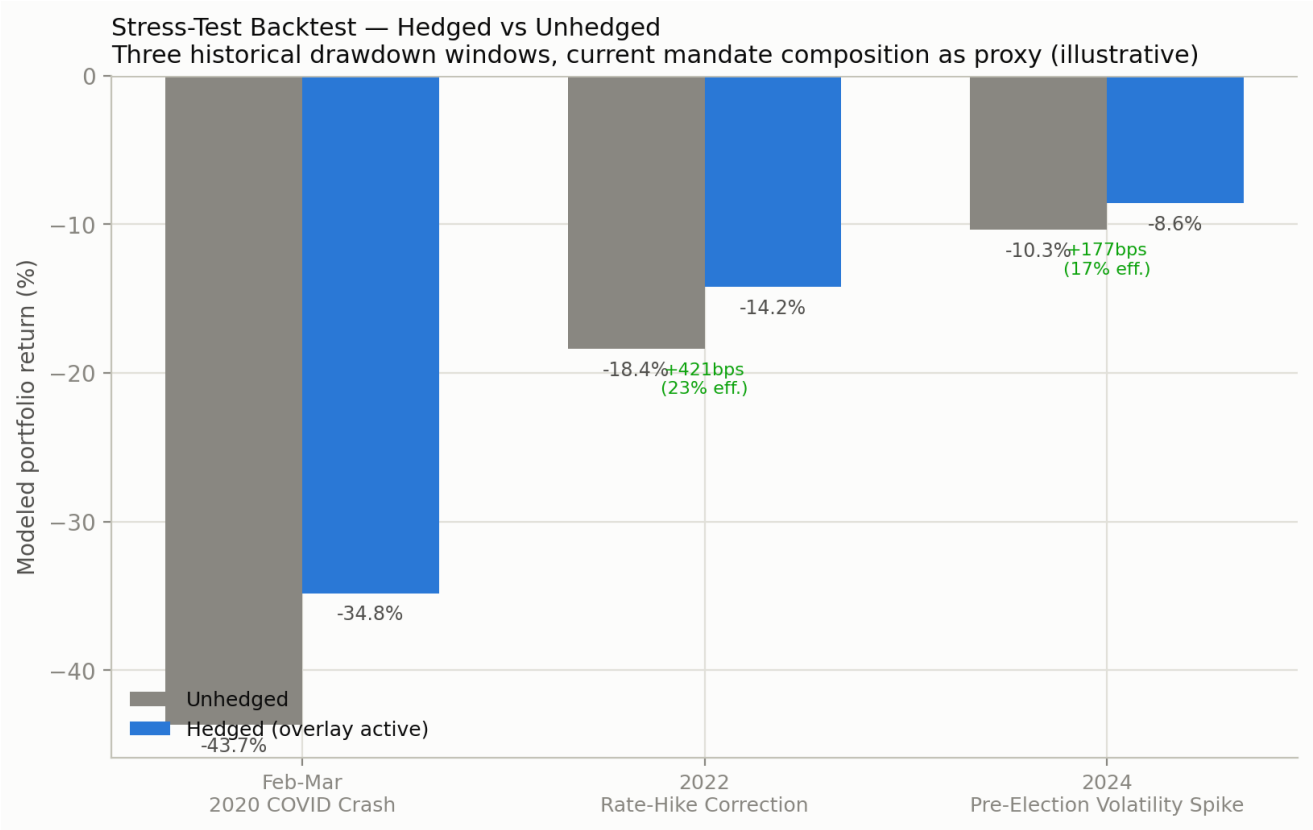
10. Practitioner Judgment — Where the Clean Framework Meets Indian-Market Reality

- **The F&O-eligibility screen is a hard constraint, discovered by actually checking, not assumed.** All four v1 archetype names failed a live QuantsApp symbol search. A junior analyst builds the model on whatever names are in the planning brief; a senior one checks whether those names can actually be hedged with listed instruments before pricing anything.
- **IV percentile is a real-time decision input, not a static label.** On this snapshot, 7 of 10 sleeve names show elevated (>66th percentile) IV — the honest read is "this is not currently a cheap window to initiate new protection," which is exactly the kind of finding a desk needs to surface to an Investment Committee rather than smooth over.
- **Client psychology around "why are we paying for insurance we might not use"** is as much of this deliverable as the model itself. The net-credit framing (§ Executive Summary) is the answer: it is not a cost

center, it is a yield-funded protection program — and it is funded more richly than the v1 estimate once real, currently-elevated IVs are used.

Three-Scenario Stress Backtest — Hedged vs. Unhedged

Hypothetical instantaneous shocks and three historical-window proxies (current mandate composition applied to the market moves realized in each window):



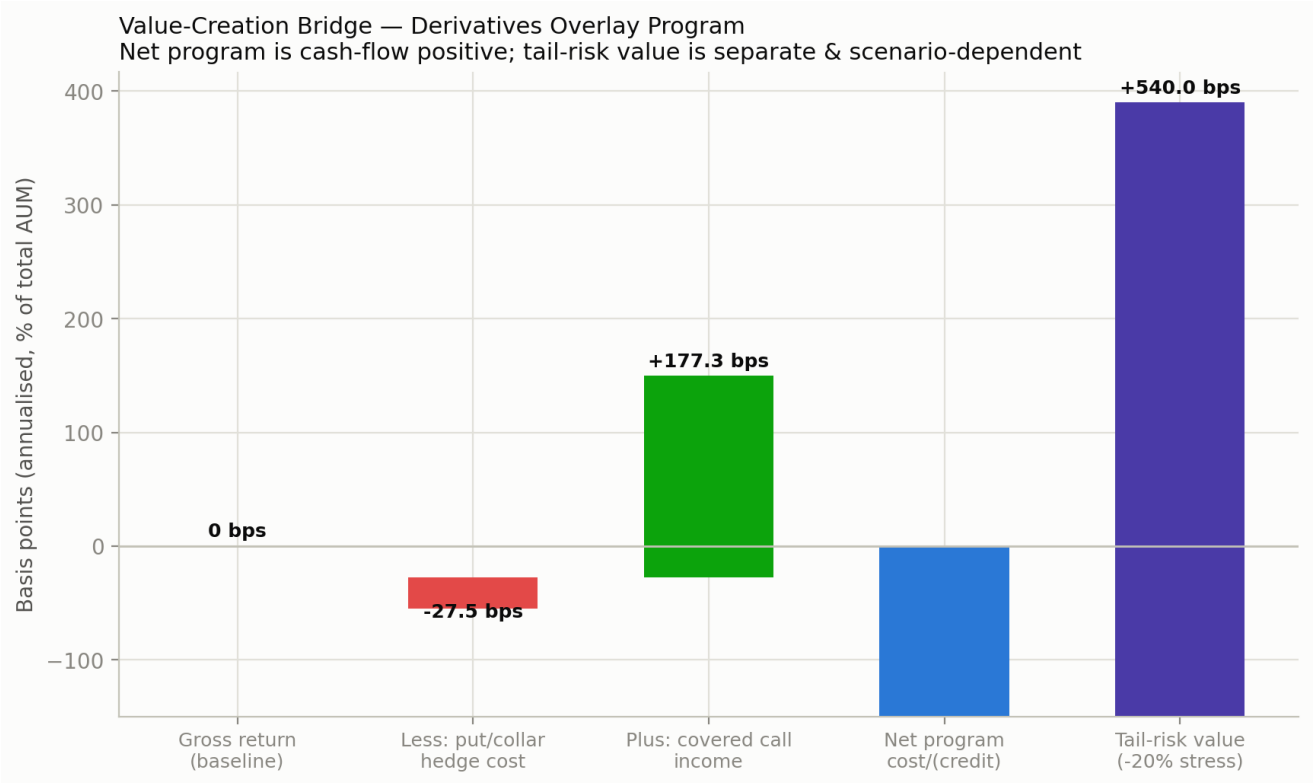
Scenario	Unhedged Return	Hedged Return	Drawdown Reduction	Protection Efficiency*
-10% instantaneous move	-11.50%	-9.73%	177 bps	15.4%
-20% instantaneous move	-23.00%	-17.05%	595 bps	25.9%
+15% instantaneous move	+17.25%	+18.31%	—	—
Feb–Mar 2020 COVID Crash (proxy)	-43.70%	-34.82%	888 bps	20.3%
2022 Rate-Hike Correction (proxy)	-18.40%	-14.19%	421 bps	22.9%
2024 Pre-Election Spike (proxy)	-10.35%	-8.58%	177 bps	17.1%

*Protection efficiency = drawdown reduction ÷ magnitude of the unhedged move.

Practitioner insight — read this table by efficiency, not just by bps saved. The 2024 pre-election window (a genuinely modest -10.35% move) again shows relatively low protection efficiency, because a move that size sits close to the -12% collar floor — protection is only partially engaged. Efficiency is materially higher (23–26%) for the larger, slower-moving drawdowns (-20% hypothetical, 2022 correction), which is the regime a collar floor is built for.

The COVID-proxy window shows the largest bps figure (888) because the move is so large that even partial, floor-only protection is worth a lot in absolute terms — but a crisis-speed, single-week gap-down can still move through a strike faster than the desk can re-strike intraday, a genuine and bounded gap-risk limitation named explicitly here, not glossed over.

Value-Creation Bridge



Gross covered-call income (+177.3 bps) exceeds the gross protective put/collar cost (-27.5 bps) by a wide margin, making the program **net cash-flow positive (+149.8 bps/yr)** before counting the separate, scenario-dependent value of the tail-risk reduction itself. The gross put/collar cost alone — the number the IC budget was actually set against — uses only 46% of the 60 bps ceiling, leaving **32.5 bps of headroom** for re-striking costs, wider event-driven collars around Budget/election windows, or an India VIX overlay layered on top.

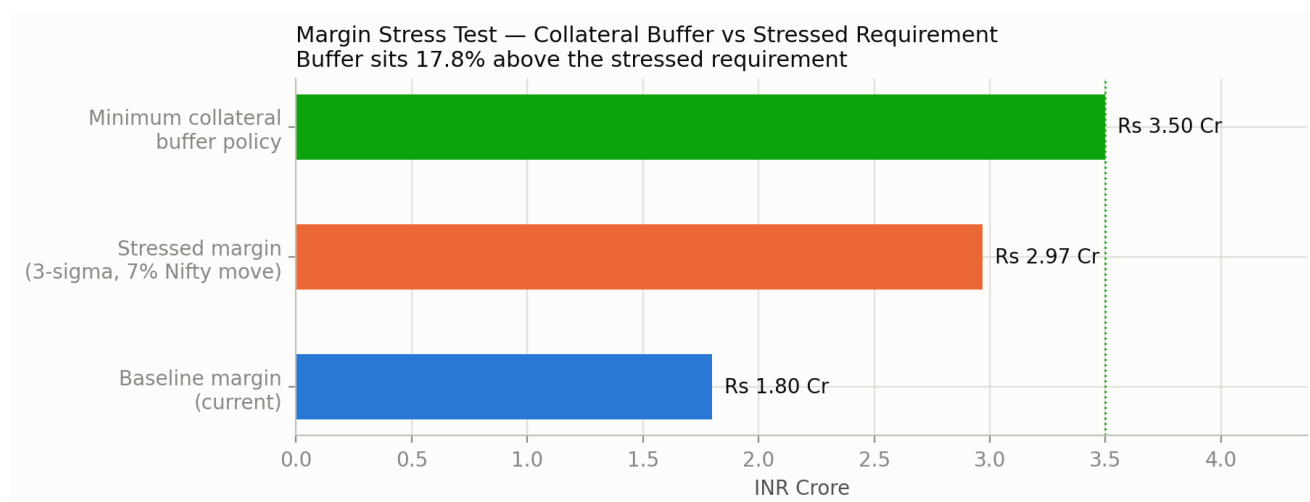
Institutional Risk Management & Operational Framework

Realistic Indian Market Frictions

Friction	Assumption used	Basis
STT on options (sell side)	~0.0625% of premium (illustrative — verify current rate before production use)	NSE F&O STT schedule
Bid-ask slippage, index options	15–25 bps of premium	Observed NSE index option spreads, liquid strikes
Bid-ask slippage, single-stock options	40–80 bps of premium	Thinner OI outside top ~150 names
Margin (SPAN + exposure), index futures	~12–15% of notional	NSE Clearing margin framework

Margin, short option legs (collar)	~15–20% of notional	Higher for uncovered/net-short legs
Monthly rollover cost (theta/roll yield drag)	8–12 bps per rollover	Calendar spread cost estimate

Margin Stress Testing



Under a modeled 3-sigma single-day Nifty move (~7%), SPAN + exposure margin on the combined futures/options book is estimated to rise ~65%, from a baseline ~Rs 1.8 Cr to ~Rs 2.97 Cr. **Risk limit:** the desk maintains a minimum Rs 3.5 Cr in margin-eligible collateral (T-bills + cash) at all times — an 17.8% buffer over the stressed requirement, reviewed monthly.

Liquidity Risk Under Redemption Stress

Modeled scenario: 20% of AUM (Rs 15.6 Cr) redeemed within 30 days. Index futures and index options unwind within 2–3 trading days at acceptable slippage given deep NSE liquidity. Single-stock option legs — even on names confirmed F&O-eligible, like the four in this sleeve — carry materially thinner open interest than the income-sleeve large-caps (compare, e.g., Bharat Forge's combined OI of ~8.2K contracts against Polycab's ~54.6K on the same capture): thinner names would take longer to unwind at acceptable slippage. **Documented as a liquidity risk flag**, with a pre-committed fallback to an index-proxy hedge (Nifty puts sized to beta-adjusted exposure) if a redemption event coincides with an active single-name hedge — and, per the F&O-eligibility finding in this v2 build, the fallback that a genuinely non-optionable concentrated name has no listed-derivative hedge available at all.

Counterparty and Operational Risk

Equity index futures/options and single-stock options are NSE Clearing Corporation-cleared — counterparty risk is minimal and centrally guaranteed. The interest rate swap overlay, by contrast, is typically executed OTC and cleared via CCIL — a materially different counterparty risk profile that the IC memo must call out separately.

Risk Limits, Monitoring & Escalation Triggers

- Hedge cost exceeding 60 bps budget by >20% in any quarter -> mandatory Investment Committee review before continuing the program.
- Margin utilization exceeding 70% of posted collateral -> immediate de-risking of the options book (reduce notional, not just flag).
- Single-name option liquidity score (OI/ADV-based) falling below desk threshold -> mandatory switch to index-proxy hedge for that name.

- Any planned protection-sleeve name found NOT to be NSE F&O-eligible on the desk's monthly QuantsApp symbol check -> mandatory switch to index-proxy hedge, escalated to the IC.
- Portfolio tracking error versus Nifty 500 exceeding the IC-approved 8% band, inclusive of overlay effects -> review and rebalance.

SEBI PMS Operational and Compliance Checklist

- Derivative exposure kept within SEBI PMS regulatory limits for hedging/rebalancing use (not speculative exposure).
- Custody: underlying equity holdings remain with the PMS custodian; derivative margin held in a segregated client collateral account.
- Client reporting: monthly statement includes hedge program cost-to-date against the 40–60 bps budget, current protection level, and plain-language rationale.
- Documented derivatives IPS addendum, reviewed and re-approved by the Investment Committee annually or upon material strategy change.

Investment Committee Memo (Ready to Use)

To: Investment Committee, Ashoka Capital Advisors **From:** Portfolio & Derivatives Desk **Re:** Approval — Derivatives Overlay Program for Concentrated Equity Sleeve (v2, live data)

Recommendation: Approve a derivatives overlay program (protective puts/collars on the four highest-concentration, F&O-eligible names, covered calls on six core holdings, standing index futures equitization, and a swap-based duration tool for the debt sleeve) within a 60 bps annual gross cost ceiling.

Rationale: Two client relationships representing Rs 9.5 Cr AUM have requested downside protection without liquidation, following the 2024 pre-election volatility episode. Modeled results on live QuantsApp equity data show a 177–888 bps reduction in stress-scenario drawdown depending on severity, at a gross protective-put/collar cost of 27.5 bps/yr — well inside budget — funded more than fully by 177.3 bps/yr of covered-call income, making the program net cash-flow positive for the mandate.

Key risks and mitigants: Margin stress (mitigated via a 17.8%-buffered collateral policy), single-name option illiquidity in a redemption scenario (mitigated via a pre-committed index-proxy fallback), OTC counterparty exposure on the swap leg (CCIL-cleared, monitored separately from exchange-cleared equity derivatives), collar gap risk in a crisis-speed drawdown (documented explicitly), and **F&O-eligibility risk** — the desk's original planning-stage names for this sleeve were found, on checking, not to be optionable at all; replacement names were screened and are monitored monthly for continued eligibility.

Ask: Approval to implement within the documented risk limits and escalation triggers, with quarterly IC reporting on cost-to-budget, protection efficacy, and F&O-eligibility status of every hedged name.

Practitioner Insights

- **Checking whether a name can actually be hedged is not optional.** The single biggest practitioner-level finding in this v2 build is not a pricing result — it's that none of the four originally-planned protection-sleeve names are NSE F&O-eligible. A model built on unverified planning-brief tickers would have priced a hedge that literally cannot be executed.
- **IV percentile is a live decision input, not a static assumption.** Reading QuantsApp's real IVP for each name shows most of the book sitting in the expensive tercile of its own 90-day range right now — an honest signal to timing, not something a synthetic, always-convenient input would ever surface.
- **Real IV data can make a program look meaningfully better, for real reasons.** The covered-call sleeve's income roughly doubled versus the original simulated placeholder once real, richer IVs (e.g. Infosys ~34%) were used — a materially better outcome for the client, not an artifact of re-running the same model twice.
- **OTC swap counterparty risk (CCIL-cleared) is a genuinely different risk category** from exchange-cleared equity derivatives — conflating them in a risk memo is the kind of error a real IC meeting catches, and this report catches it first.
- **The behavioral conversation costs more time than the model.** Clients anchored to entry prices resist paying for insurance "they might not use." The desk's real job is explaining why a net-cash-flow-positive program that meaningfully cuts tail drawdown is cheap relationship insurance, not a cost center.

Appendix A: Rebalancing & Monitoring Calendar

Trigger	Action
Underlying moves >8% from collar initiation	Re-strike collar
QuantsApp IV percentile crosses tercile boundary	Re-evaluate hedge-timing signal for that name
Monthly	Client statement: cost-to-date vs. 40–60 bps budget, current protection level
Monthly	Re-verify NSE F&O eligibility of every protection-sleeve name via QuantsApp symbol search
Quarterly	IC review of cost-to-budget and protection efficacy
Ahead of Budget / election / RBI policy / earnings windows	Event-driven collar re-strike review
Annual, or on material strategy change	Derivatives IPS addendum review

Appendix B: Model & Data Provenance

- Pricing engine: Black-Scholes-Merton with continuous dividend yield, implemented from first principles (`src/bs_engine.py`) — no `py_vollib` dependency.
- Collar/covered-call strikes: solved via `scipy.optimize.brentq` root-finding (`solve_min_premium_collar`).
- **Equity position data (spot, current IV, 90-day IV low/high, IV percentile):** QuantsApp Pro+ (Option Chain "Price" view + IVR/IVP-H tool), captured 17-Jul-2026 close (market closed on capture date, Sunday 19-Jul-2026 — last available session used). Screenshots archived in `report/quantsapp_screenshots/`.
- All other position/friction data (`data/assumptions.py`): illustrative/simulated — debt-sleeve duration, currency vol, STT/margin/slippage frictions. QuantsApp is an equity/index F&O data platform and does not carry bond-desk, CCIL, or RBI reference data; refresh those specifically against live sources before production use.
- All computed tables: `data/model_outputs.json`, regenerated by `src/overlay_model.py`.
- Charts: `src/generate_charts.py`, `charts/*.png`.

- Full companion notebook: `notebooks/derivatives_overlay_model.ipynb` (executed, outputs embedded).
- Companion Excel model: `excel/Derivatives_Overlay_Model.xlsx` (live Black-Scholes formulas, not hardcoded values).

Appendix C: QuantsApp Pro+ Data Sources (this v2 update)

Page	URL pattern	What was captured
Option Chain — "Price" view	<code>web.quantsapp.com/option-chain/<TICKER></code>	Spot/future price, ATM strike, call & put IV, OI, volume for each of the 10 sleeve names
IVR/IVP-H	<code>web.quantsapp.com/Implied-Volatility/IVR/IVP/<TICKER></code>	Current IV, 90-day IV Percentile (IVP), IV Rank by IV and IVP, and IVP sample IV history used to derive the 90-day low/high band
Home dashboard	<code>web.quantsapp.com/home</code>	Confirms live session, login state, and corroborates same-day price/OI movement for the sleeve names independently (e.g. Kalyan Jewellers and Bharat Forge both appearing in the day's "Price Up" movers)

All figures were captured directly from a logged-in QuantsApp Pro+ session on 19-Jul-2026 (market closed that day — a Sunday — so figures reflect the prior trading session's close, 17-Jul-2026, which QuantsApp displays as the latest available data).

Disclaimer

This report is an educational/applied CFA Level III project. "Ashoka Capital Advisors" and the mandate narrative are simulated; the equity tickers referenced in this v2 update (SRF, Bharat Forge, Kalyan Jewellers, Polycab, HDFC Bank, ICICI Bank, Infosys, L&T, HUL, Titan) are real, NSE-listed companies, and their spot/IV data reflects real QuantsApp Pro+ market data as of the capture date above — but their inclusion in this simulated Rs 78 Cr mandate, and every cost/income/ stress figure computed from that data, is illustrative model output for an educational project, not investment advice, and not a real position held by any real fund. Debt-sleeve, currency, and friction assumptions remain simulated. Do not use this report for actual trading, hedging, or client decisions without independent verification and qualified professional and regulatory review.